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Tidal stirring and its impact on water column stability and property distributions in a semi-enclosed shelf sea (Seto Inland Sea, Japan)

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Abstract

The Seto Inland Sea is a partially enclosed shallow sea with a complex bathymetry involving several basins, separated by narrow straits, and numerous islands. The influence of tidal stirring on density stratification and on the distribution of nutrients and chlorophyll in this region has been investigated using a combination of a numerical model of the tidal flow and the results of two extensive surveys of property distributions in the summer regime. The Princeton Ocean Model (POM) has been used to determine the spatial structure of the tidal flow and the results have been verified by comparison with an extensive database of current and elevation measurements. The model indicates a large difference in total energy flux between the western entrance section, which is controlled by a progressive wave regime, and the eastern entrance section which is associated with a predominantly standing wave regime. The combined energy flux input matches the total energy dissipation for the M_2 in the Inland Sea which is found to be $\sim 3.4 \times 10^9$ W. Two ship surveys in August 2002 and 2003 along a central section show a highly differentiated pattern of vertical structure. The potential energy anomaly φ, an index of vertical stability based on density, exhibits a marked inverse correlation with tidal energy dissipation $\overline{D}$ thus confirming the control of stratification by tidal stirring. In most straits, vertical differences of nutrients and chlorophyll are small or negligible while in the low-energy basins there are pronounced vertical differences associated with density stratification. These sluggish basins exhibit oligotrophic conditions in the surface layer but large nitrate concentrations below the pycnocline and significant chlorophyll maxima in subsurface layer. The physical pattern of varying stratification and fronts is reflected to a considerable degree in the distributions of biochemical properties.
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1. Introduction

Tidal motions on the shelf induce strong frictional stresses at the seabed, resulting in vigorous turbulent stirring. This stirring, which has a spatial distribution controlled by tidal dynamics, competes

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with the buoyancy inputs from surface heating and freshwater to determine the pattern of density stratification in tidally energetic shelf seas (Simpson, 1998).

The Seto Inland Sea is a semi-enclosed shelf sea connected to the Pacific Ocean through two openings with the distance along the channel being ~550 km. The complicated geometry involves the widths across the channel varying from 0.5 to 60 km and creates a series of broad basins and narrow straits. Tidal currents are intensified in those narrow straits and reach speeds up to 5 m/s (Takeoka, 2002). Exchange flows arising on the shelf when a pronounced pycnocline develops (Millot, 1990), may carry oceanic nutrients into shelf seas and consequently affect water quality within the shelf sea (Jickells, 1998). There is evidence that significant amounts of oceanic nutrients intrude in this way from ocean into the Seto Inland Sea (Fujiwara et al., 1997). If the oceanic nutrients are supplied to the euphotic zone, they contribute to the primary production. The distribution of tidal stirring, and hence vertical mixing, is an important factor controlling the pathways of these nutrients.

The distribution of tidal energy dissipation in the Seto Inland Sea can be conveniently represented by the h/u^3 parameter defined by Simpson and Hunter (1974) as reported by Yanagi and Okada (1993) and Takeoka (1990). Using satellite data, CTD measurements, and numerical model, Takeoka (1997) and Yanagi and Tamaru (1990) investigated the tidal fronts in the western opening, and proposed that the location of tidal fronts could be predicted by the h/u^3 parameter.

In recent years, improving measurement techniques have been providing fine and large-scale information on the distribution of turbulent mixing in shelf seas (e.g. Gargett, 1999; Rippeth et al., 2003), and the results are found to be reasonably well reproduced by models using a turbulent closure scheme (Simpson et al., 1996). Total turbulent energy dissipation estimated by modeling has been found to be in good agreement with total tidal energy input (Davies and Kwong, 2000).

In this study, we investigate the distribution of the tidal currents and the associated turbulent dissipation using a numerical model over the whole area of the Seto Inland Sea in order to determine the tidal energy flux and the distribution of tidal stirring. We then use the results of the model to interpret extensive surveys of the vertical distribution of physical and biochemical properties made in the summer regime of 2002 and 2003 to investigate the links between tidal stirring, water column stratification and the vertical distribution of nutrients and chlorophyll pigments.

2. Distribution of tidal currents and mixing

2.1. Model condition and computation of parameters

A numerical model, based on the Princeton Ocean Model (POM) (Blumberg and Mellor, 1987) and including a second moment turbulence closure scheme (Mellor and Yamada, 1982), has been applied to determine the distribution of tidal currents and energy dissipation. Since the aim was to model the tidal current system, the model was forced only by tidal elevation and does not include density variations and buoyancy inputs at the surface or from rivers.

The bathymetry and model domain are shown in Fig. 1a and b. The horizontal grid size is 2 km square while eight non-uniform sigma levels are used in the vertical. The depth data were obtained by smoothing from a fine resolution (500 m) bathymetry supplied by Japan Ocean Data Center (JODC). Because the aim of the model is to determine the distribution of tidal current and tidal stirring in the Seto Inland Sea, we assumed that baroclinic effects were negligible and density was set constant following the approach used in 2-d models of tidal streams (e.g. Pingree and Griffiths, 1978; Garrett et al., 1978).

There are three openings in the Seto Inland Sea. One of them, the Kanmon Strait at northwestern boundary, is assumed to be closed (shown by solid line in Fig. 1b). Since the channel is very narrow (~0.6 km), water exchange across the boundary is small and its effects on the distribution of the tidal currents in the Seto Inland Sea are minimal (Hibino et al., 2000; Takeoka, 2002). The model was forced by the dominant M_2 tidal elevation, along the two open boundaries (shown by broken line in Fig. 1b). The tide data is derived from harmonic constants data provided by the Maritime Safety Agency.

Tidal energy flux vector at each point over an M_2 period T is calculated from the following defining equation (Greenberg, 1979):

$$(E_i, E_j) = \frac{1}{T}\int_0^T (U, V)\rho h\left\{g\eta + \frac{1}{2}(U^2 + V^2)\right\}\mathrm{d}t,$$

where U and V are depth-averaged tidal velocities, ρ is density of this region, h is the water depth, g is of

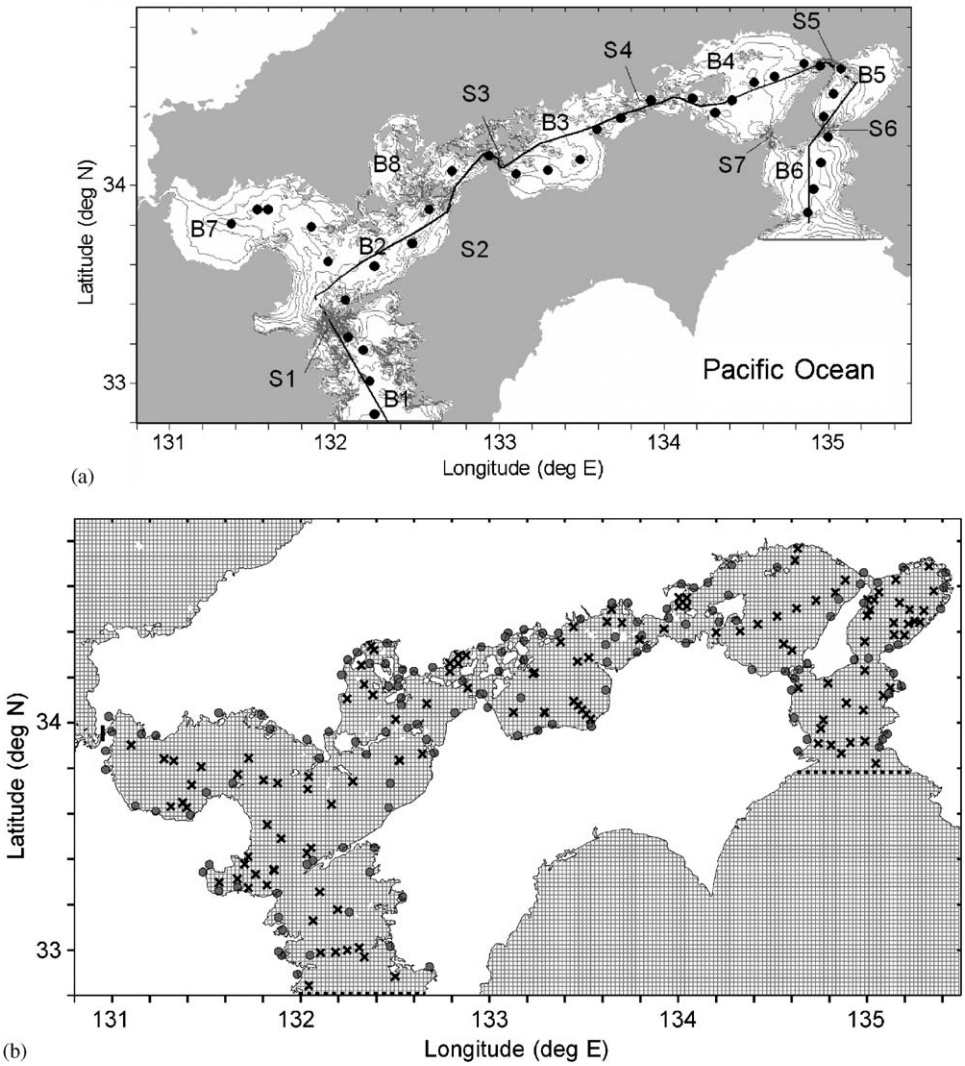

Fig. 1. (a) Bathymetry of the Seto Inland Sea. The contours indicate water depth in meters (contour interval = 10 m). Circles are observation points and bold line is the axial line of the channel. The points indicated by black circles will be reflected on the line and used for axial sections. (b) Grid of the three-dimensional model. A solid line indicates closed boundary and broken lines indicate open boundaries. Circles show the location of tide gauge and crosses are the positions of current meters observations used for validation of the model.

gravitational acceleration and η is displacement of the sea surface from the equilibrium level.

The mean rate of energy dissipation rate per unit volume over M_2 period T is determined, following Davies and Kwong (2000):

$$D = \frac{1}{h}\frac{1}{T}\int_0^T \int_0^N \left[\frac{u\,\partial\tau_i}{\partial\sigma} + \frac{v\,\partial\tau_j}{\partial\sigma}\right] \mathrm{d}\sigma\,\mathrm{d}t,$$

where N is number of sigma layer, h is water depth. τ_i, τ_j are the shear stress component given by

$$\tau_i = -\frac{\rho A_\mathrm{v}}{h}\frac{\partial u}{\partial\sigma} \quad \tau_j = -\frac{\rho A_\mathrm{v}}{h}\frac{\partial v}{\partial\sigma},$$

where ρ is density and A_v is vertical eddy viscosity derived from turbulence closure scheme.

The stress component at the bed is computed from

$$\tau_i = C_\mathrm{D}\rho u_\mathrm{h}(u_\mathrm{h}^2 + v_\mathrm{h}^2)^{1/2} \quad \tau_j = C_\mathrm{D}\rho v_\mathrm{h}(u_\mathrm{h}^2 + v_\mathrm{h}^2)^{1/2},$$

where $u_\mathrm{h}v_\mathrm{h}$ are the currents at bottom layer and C_D is the coefficient of bottom friction used in the model.

The drag coefficient was set to the larger of 0.0025 and

$$C_\mathrm{d}\left\{\frac{\kappa}{\ln(h/z_0)}\right\}^2,$$

where κ is Karman constant, z_0 is the roughness height, and h is the thickness of bottom layer in each mesh. A simple empirical relationship $z_0 = 0.01 + 1/H$ (where H is depth) is used to allow for the increase of bottom roughness in shallow water.

2.2. The model results

2.2.1. Validation of model results

The modeled harmonic constants of the M_2 tides were validated by a comparison with those derived from tidal elevations based on 130 tide gauges. The positions of observation points are shown in Fig. 1b. The correlation coefficients for amplitude and phase were 0.978 and 0.984 ($n = 128$), respectively, indicating satisfactory agreement between model and observations.

The modeled tidal currents are similarly validated by a comparison to the data derived from 189 current time series (locations in Fig. 1b) measured by the Maritime Safety Agency and accumulated by National Institute for Industry. Fig. 2 shows the correlation between observed and calculated amplitude of major axis of M_2 tidal current. Although agreement is generally good, there was relatively poor correlation at a few sites. Inspection revealed that these sites are located, in complicated topography such as around straits and between islands or in highly irregular coastlines. After removing those data at such sites where spatial differences of tidal currents are too large to allow meaningful comparison, the correlation coefficient is 0.71 and the slope is 1.03.

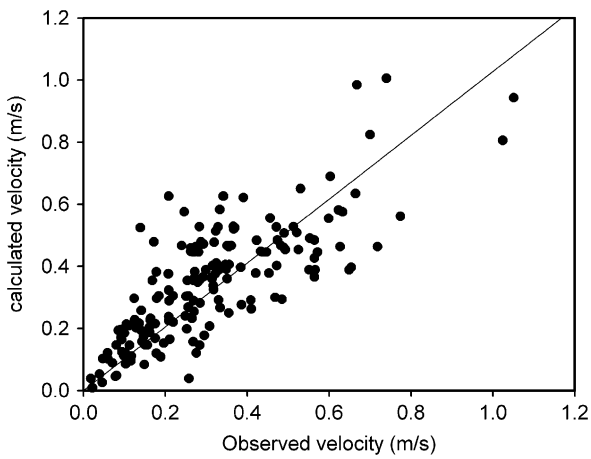

Fig. 2. Relationship between calculated and observed amplitudes of M_2 tidal current.

The root-mean-square errors between model results and observations for the amplitude of tidal displacement and major axis of tidal currents by M_2 component over the entire Seas are 4.5 cm, and 0.11 m s^{-1}, respectively.

2.2.2. Tidal currents and energy flux

Fig. 3a shows the distribution of tidal ellipses. Those at straits (S1–S7) and in the regions close to coast are almost rectilinear. The amplitude of the tidal current is generally larger in the western part (<133°E) and smaller in eastern part (>134°E). Tidal currents at straits are typically stronger than those in basins, but the contrast is especially evident in basins B3, B4 located in inner part of the Sea. The currents in the regions close to coast and in the embayments outside the main path of the tidal wave are also weak.

Fig. 3b is the distribution of tidal energy flux vector. It shows a large energy input from the western opening. Most of this flux goes towards the east along the main channel and the eastward flux continues through to strait S4. The net energy flux input from the eastern opening is relatively small, and it decreases rapidly towards the west, becoming negligible when it reaches basin B4.

Fig. 3a and b show that the tidal currents and the energy flux in the basin (B7) diminish towards the northwest as they must because of the assumption of the Kanmon strait being closed in the model. In reality, there is some exchange through this channel but in view of the small cross-section of this entrance any effect on the energy distribution will be localized and will not have a significant effect on the overall distribution of tidal energy in the Seto Inland Sea.

2.2.3. Tidal energy dissipation

Fig. 3c shows the distribution of mean rate of energy dissipation averaged over M_2 tidal period ($\overline{D}$, W m^{-3}). The contour lines are drawn for $\log_{10}\overline{D}$. Estimated dissipation in western part (<133°E) is generally 10 times as large as that in eastern part (>134°E), reflecting larger tidal currents in western part. Regions with high rates of dissipation are found in most straits and their vicinities (S1–S7). The large dissipation in straits S5–S7 is responsible for the rapid decrease in the energy flux from the eastern openings into basin B4. In contrast to the high-energy loss in the straits, dissipation is generally low in basins and is especially low in basin B4.

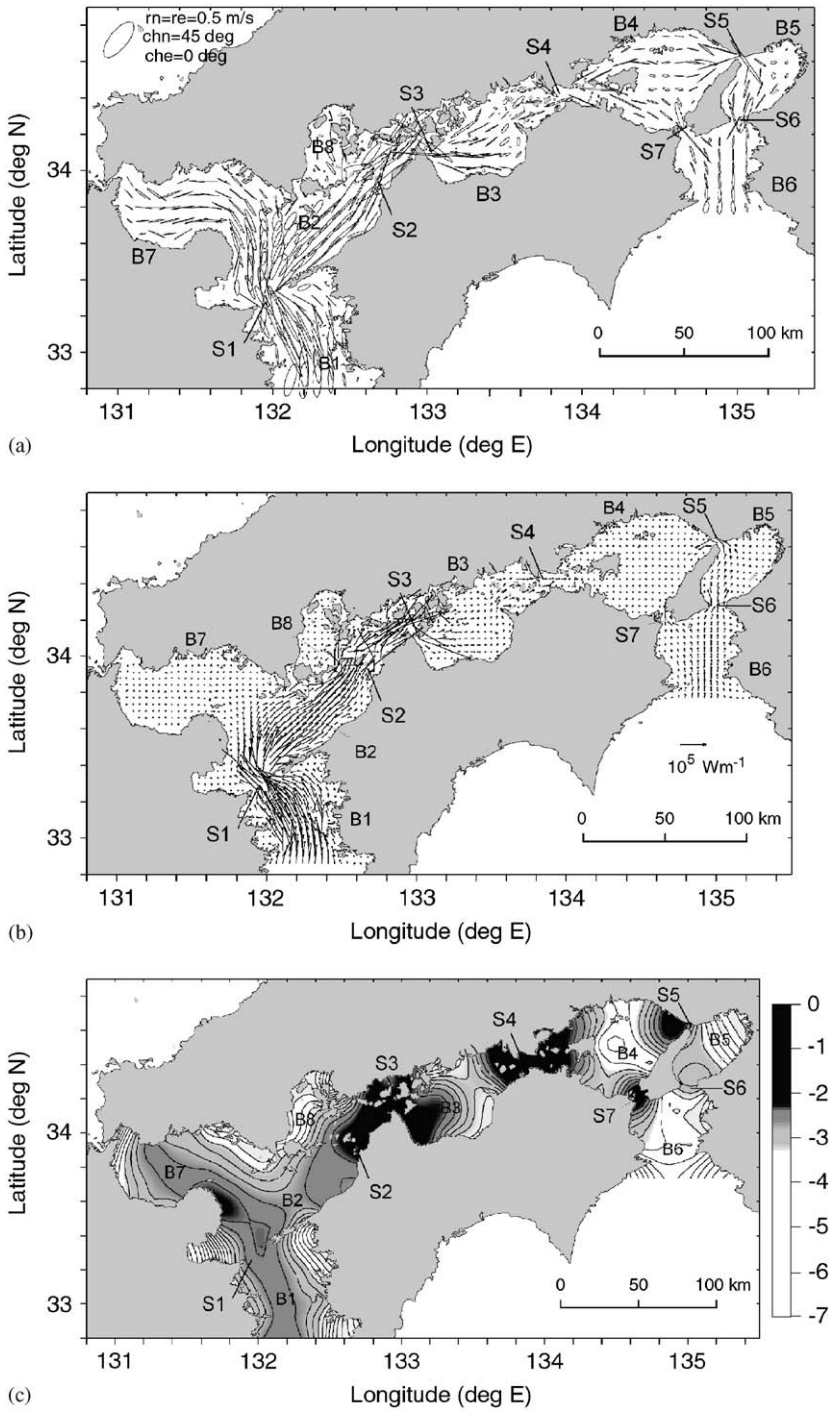

Fig. 3. Calculated horizontal distributions of (a) tidal ellipses, (b) tidal energy fluxes ($10^6\,\mathrm{W\,m^{-1}}$) and (c) mean dissipation ($\overline{D}$, $\mathrm{W\,m^{-3}}$). Arrows and symbols indicate features referred to in the text.

2.2.4. Energy budget

The total energy input by the M_2 tide from western and eastern boundaries is computed by integrating the energy flux at each section. The results from western and eastern part are estimated as 2.9×10^9 and 0.6×10^9 W, respectively. As shown in Fig. 3b, the majority of tidal energy flux comes from western part. On the other hand, total energy

dissipation over the whole region is estimated as 3.4×10^9 W, which is in close accord with the total energy input into the Inland Sea.

The result shows asymmetry of energy input with about 5 times more energy propagating through the western opening than through the eastern opening. Tidal waves propagating from Pacific Ocean through both openings meet at the central area around 134°E. To the west of this position, the tide is predominantly a progressive wave, while in the eastern part there is a standing wave as noted by Takeoka (2002). The contrast in energy input between western and eastern parts is mainly due to the different form of the tidal wave and the significantly greater water depth, and, hence, cross-sectional area in the western entrance.

3. Water column stability and property distributions

3.1. Observations

The first survey was undertaken during the period from 1st to 5th August 2002 and the second from 4th to 5th August 2003. The total number of stations was 34 extending over the whole of the area (Fig. 1a). In these cruises, vertical profiles of temperature and salinity were measured and water samples were taken at all these stations, at depth intervals of 5–10 m. At the same time, measurements were made of fluorescence, along with light intensity. Chlorophyll-*a* concentration in discrete samples was determined by Turner fluorometer and these results were used to convert the fluorescence to the values of chlorophyll *a*. Nutrients (phosphate, nitrate) were analyzed using an AutoAnalyzer II, while particulate nitrogen and carbon were analyzed using a Yanako CHN coder.

3.2. Distribution of physical properties

3.2.1. Temperature

The vertical distribution of observed temperature is shown in Fig. 4a. Pronounced vertical gradients of temperature occur in almost all of the basins. Cold domes are formed in the bottom of basins B2 and B4. Cold-water masses are also found on the bottom of both openings to the ocean (B1, B6). By contrast, the temperature profiles at stations located in, or near to, straits are vertically homogeneous on nearly so. At strait S3, conditions are completely

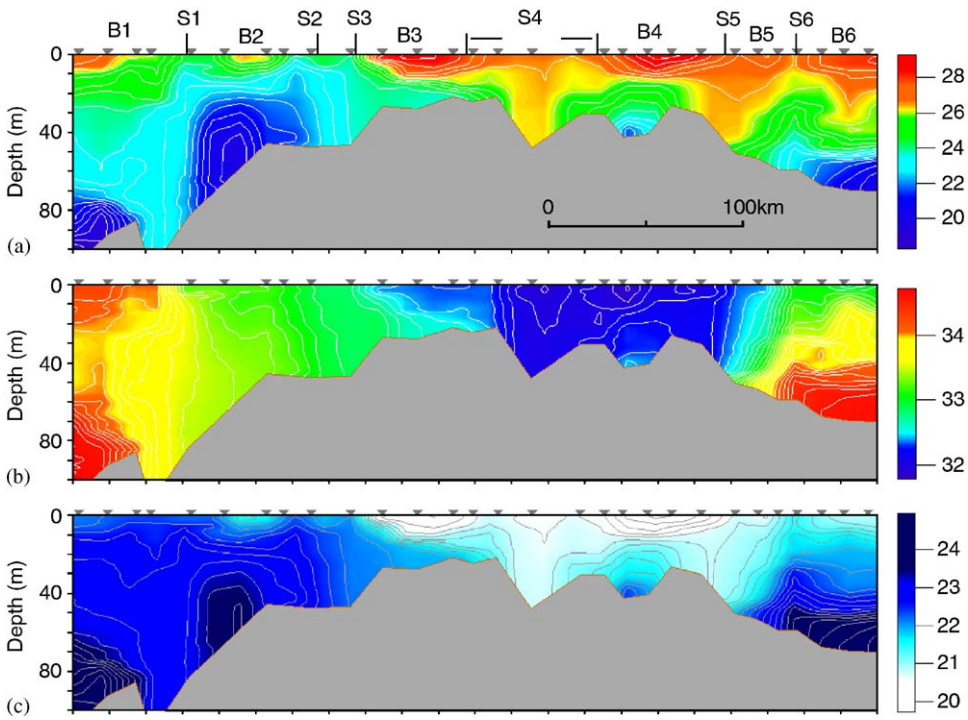

Fig. 4. Axial sections of (a) temperature (°C), (b) salinity (psu) and (c) sigma-t observed in August 2002. Triangles above each figure indicate observation points. The location of the stations along the axial line is shown in Fig. 1a.

mixed while those near S2, S5 and in the center of S4 are nearly homogeneous. At strait stations near S1 and S6, where the narrows are relatively short, vertical gradients remain.

Surface temperatures at stations located at, or near, straits are typically substantially lower than those in basins. This causes pronounced horizontal gradients between straits and basins, which take the form of surface and bottom fronts. In the frontal regions, typical gradients are $\sim 0.1\,^{\circ}\mathrm{C\,km^{-1}}$ while the maximum observed horizontal gradient occurred in 133°E with the value of $0.2\,^{\circ}\mathrm{C\,km^{-1}}$.

3.2.2. Salinity

In comparison to the temperature, vertical gradients in salinity were relatively weak (Fig. 4b). On the bottom of western and eastern openings (B1, B6), cold saline water indicates the inflow of oceanic water. Saline water is also apparent in the surface layer of western opening (B1), suggesting the surface intrusion of warm saline oceanic water.

There is a minimum in salinity of 31.4 in the strait S4 (134.2°E) with salinity increasing outward to both western and eastern entrance where S∼34.5. While saline water is apparent in the surface layer of western opening (B1), isohalines in both openings are generally tilted upwards towards the open sea.

3.2.3. Density

The density field derived from observed temperature and salinity is shown in Fig. 4c. As with the salinity, the density increases seawards from a minimum centered on the strait S4 to maximum values at the ocean boundaries. Like the isohalines, the isopycnals in both openings slope upwards towards the open sea, suggesting a component of estuarine circulation with outflow of light water in the upper layer and inflow of heavy water in the lower layer. The general pattern of the vertical structure of temperature, i.e. stratified in basins and vertically homogeneous in straits, is reflected in the vertical structure of density with frontal boundaries in density between basins and straits.

3.3. Water column stability and its controlling factor

As an index of overall water column stability, we use the potential energy anomaly φ ($\mathrm{J\,m^{-3}}$) representing the amount of work per volume required to maintain complete vertical mixing. φ is calculated from the density profile according to Simpson and Bowers (1981);

$$\varphi = \frac{1}{h}\int_{-h}^{0}(\overline{\rho}-\rho)gz\,\mathrm{d}z \quad \text{where} \quad \overline{\rho} = \frac{1}{h}\int_{-h}^{0}\rho(z)\,\mathrm{d}z.$$

Here g is gravitational acceleration ($\mathrm{m\,s^{-2}}$), ρ is water density ($\mathrm{kg\,m^{-3}}$), h is water depth (m), z is depth coordinate (m).

The broad distribution of observed water column stability can be seen in Fig. 5a, $\log_{10}\varphi$ at each station. The potential energy anomaly in basins was typically 10 times as much as those in straits. Moreover, the potential energy in the deeper water of both openings (<132 and >135°E) is around 2.5 times as much as in basins located in the inner region. This, in part, reflects the influence of deep ocean stratification intruding into the shelf sea. In contrast, the values of $\log_{10}\varphi$ at straits are typically less than 1.5, although at stations near some straits, for example S1, S5, S6, relatively high stability occurs because the parameter φ also includes the positive input by fresh water discharge as well as heating.

Fig. 5b–d show the longitudinal distributions of mean tidal energy dissipation (model result), magnitude of M_2 tidal current (model result validated by data) and channel width, respectively. The width of the channel is defined along the direction of the normal to the axial line (see Fig. 1a). In narrow regions, tidal currents are generally intensified, although it should be noted that there are large spatial variations of velocity in strait S4, reflecting the rather complicated bottom topography and horizontal geometry. The magnitude of the mean dissipation in straits is up to ∼100 times as large as that in basins.

The longitudinal distribution of water column stability has an inverse correlation with energy dissipation. A regression of $\log_{10}\varphi$ versus $\log_{10}\overline{D}$ explains 42% of the variance with 25 degrees of freedom, a result, which is significant at the 0.1% level. Stratification is strongest in basin B4 corresponding to the lowest dissipation. It is also strong in basin B3 where again dissipation level is low. In contrast, stratification is weakest in the region between straits S2 and S3 where the dissipation level is highest.

3.4. Distribution of biochemical properties

3.4.1. Nutrients

Fig. 6a shows the observed distribution of nitrate concentration. The dotted line in this figure indicates

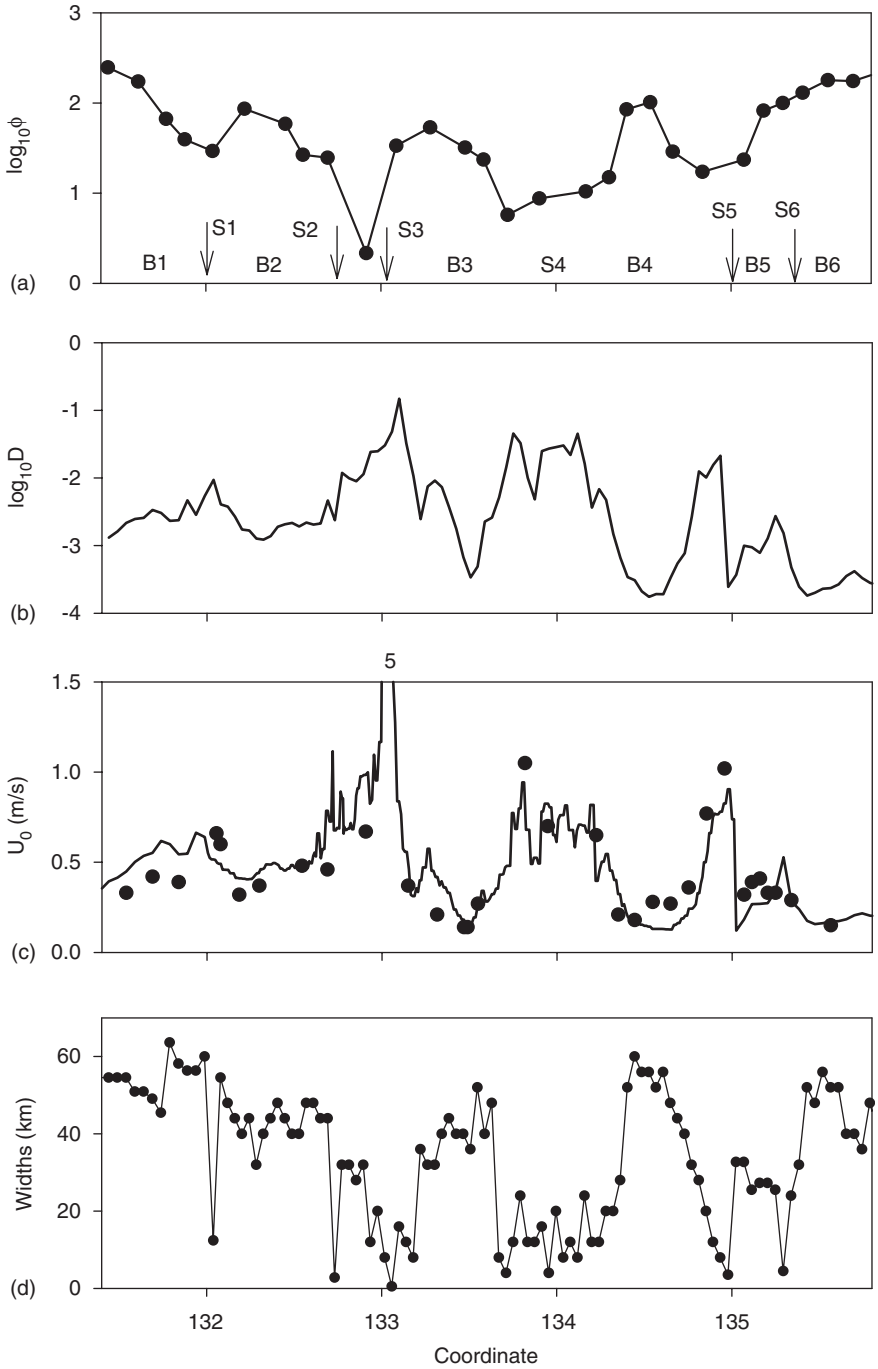

Fig. 5. Longitudinal distributions along the axial line (see Fig. 1a) of (a) potential energy anomaly at each observation points (φ, J m^{-3}), (b) mean dissipation ($\overline{D}$, W m^{-3}), (c) tidal current (m s^{-1}) and (d) width of the channel (km). The lines in (b) and (c) are the model results extracted every 2 km and circles in (c) are observed tidal currents (see Section 2.1).

the compensation depth. This is the lower limit of the euphotic zone and is estimated from the observed light intensity, as the depth at which the light intensity is 1% of its value at the sea surface at the sea surface. In basins such as B2, B3, B6 and especially B4, the concentrations of nitrate observed in the upper layers were near the detection limit. At the same time, much higher levels of nitrate occurred

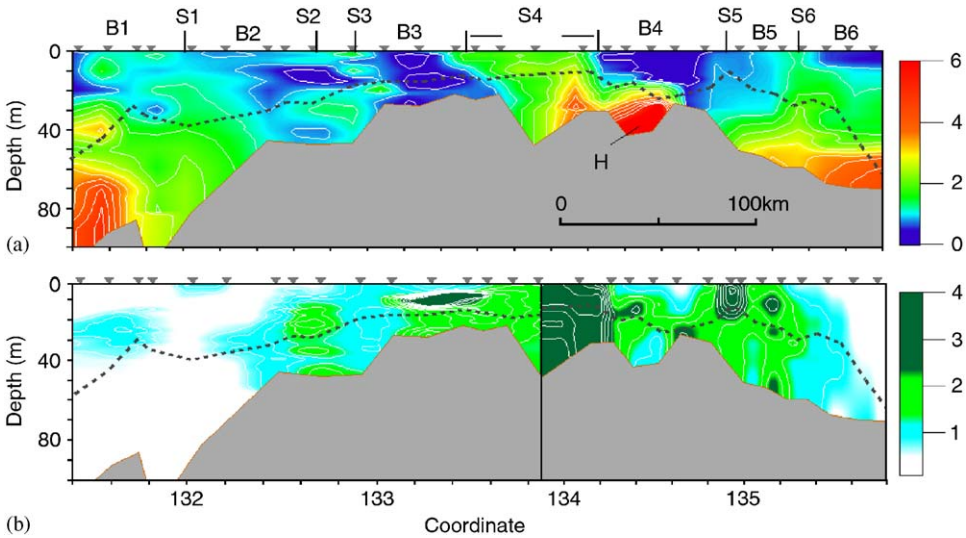

Fig. 6. Axial sections of (a) nitrate (μM) observed in August 2002, and (b) chlorophyll-*a* ($\mu g\,l^{-1}$) observed in summer (left figure is in August 2002 and right figure is in August 2003). Broken line in (a) is the compensation depth. The location of the stations along the axial line is shown in Fig. 1a.

in the deep water of the basin B4 and the western and eastern openings B1, B6. In basin B4, concentrations of up to 13.0 μM were found in the deepest sample which was from isolated, cold water with slightly enhanced salinity (Fig. 4a, b). This water mass has a low oxygen concentration of around 30% saturation, which suggests the operation of recycling processes of organic matter increasing nutrient concentrations. The highest concentrations in basins B1, B6 were 5.0 and 4.0 μM, respectively, values which corresponded to cold and saline water intruded from the open sea.

At most of stations located at or near straits, concentrations of nitrate were more or less vertically homogeneous and those at the surface were relatively higher than surface values in the basins. Concentrations of nitrate were homogeneous and high in strait S4, presumably because of upward mixing of nutrient rich water from the adjacent Basin B4.

3.4.2. Chlorophyll

The distribution of chlorophyll concentration, which is an indicator of phytoplankton biomass, is shown in Fig. 6b. In the western part (<133°E), concentrations are generally lower than in the central and eastern part of the Seto Inland Sea (>134°E) and the majority of concentration values were lower than 0.4 μM, which reflects oligotrophic conditions in the western part.

In several of the basins, there is evidence of chlorophyll maxima layer occurring in the pycnocline of the stratified water column, which generally lies above the compensation depth. In basin B3, for example, a pronounced mid-water maximum extends right across the basin with concentrations of the subsurface peaks up to 5.0 $\mu g\,l^{-1}$. In contrast, at the stations in straits S4 and in the region between straits S2 and S3, chlorophyll is homogeneously distributed in the water column.

3.4.3. The longitudinal distribution of depth-mean biochemical properties

In addition to the vertical structure imposed by tidal stirring, there are also significant horizontal gradients in the biochemical distributions. Fig. 7a shows the longitudinal distributions of depth mean concentration of nitrate (μM) within euphotic layer and in lower layer. There are high values of nitrate in the lower layer of western and eastern opening and these peaks gradually decrease toward the interior of the Inland Sea. Nitrate concentration rises again in the region 133.5–135°E with a pronounced maximum in the lower layer of basin B4. The inference seems to be that nutrient fluxes from the ocean are supplemented by run-off sources and re-cycling in the interior near-bed region.

Fig. 7b shows the longitudinal distributions of the mean concentration of chlorophyll ($\mu g\,l^{-1}$) within the euphotic layer. It shows two significant peaks at 134°E (S4) and 135°E (S5), which are located on the boundaries of basin B4. From these peaks, the mean concentration gradually decreases toward both

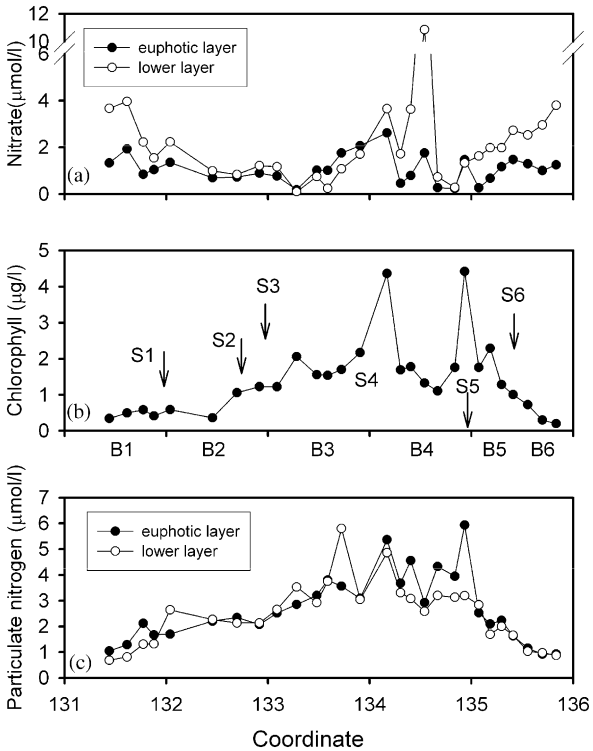

Fig. 7. Longitudinal distributions of (a) mean concentration of nitrate in euphotic layer (closed circles) and in lower layer (open circles) (μM), (b) mean concentration of chlorophyll in euphotic layer ($\mu g\,l^{-1}$), (c) mean concentration of particulate nitrogen in euphotic layer (closed circles) and in lower layer (open circles) (μM). The location of the stations along the axial line is shown in Fig. 1a.

openings. The mean concentration of particulate nitrogen (Fig. 7c) exhibits a rather similar spatial pattern to the distribution of chlorophyll, suggesting nitrate (dissolved nitrogen) is converted into particulate nitrogen through nutrient consumption by phytoplankton. The high concentrations of particulate nitrogen in the lower layers of basin B3 is suggestive of a flux into the bottom layer due to the sinking of particles.

4. Summary and discussion

In this study, we have examined the relation between the barotropic tidal flow field, the associated flux of tidal energy and their impact on the observed distribution of stratification and the biogeochemical parameters. The main results are (i) the determination of the fluxes of tidal energy into the system, (ii) the elucidation of the control by tidal stirring of stratification of the temperature, salinity and density distributions and (iii) the response of the biogeochemical distributions to variations in the physical structure. In all these aspects, we consider that we have advanced on the pioneering work of Yanagi and Okada (1993) (reviewed in Takeoka (2002)).

The energy input by M_2 tide to the Seto Inland Sea ($\sim 3.5 \times 10^9$ W) is dissipated during the propagation of the tidal wave from the two entrances to a null flux region in the interior. The divergence of this energy flux supplies the energy dissipation, which exhibits large spatial variability. This variability is reflected in the pronounced contrast of properties between the different regions of the inland sea. The distribution of water column stability exhibits a marked inverse correlation with tidal energy dissipation (Fig. 5a, 5b) in spite of imperfect spatial matching of the two fields. This latter problem may be attributed to the sampling interval being too large to resolve the high gradient regions and to water column displacements by the semi-diurnal tide.

The effect of differences in stability was clearly apparent in the distribution of biochemical properties. In most straits, vertical differences of nutrients and chlorophyll were small or negligible while in the low-energy basins there were pronounced vertical differences associated with density stratification. Cold domes existed on the bottom of all basins and some of these contained high concentration of nutrients (Fig. 6a). In most basins, sub-surface chlorophyll maxima were formed, whereas concentrations in the immediate surface layer were nearly zero. The inference is that the level of tidal mixing is exerting the key control on the vertical gradients of biochemical properties.

Distributions of temperature, salinity and nutrient suggested intrusion of oceanic water thorough adjacent shelf (Fig. 4a, b, 6a). Intrusion of cold and nutrient rich water masses from lower layer of shelf water have been reported in most years (e.g. Fujiwara et al., 1997). The high salinity occurring in a surface layer at the west boundary (Fig. 4b) is also considered to indicate the intrusion of shelf water. It is called 'Kyucho', a phenomena of intrusion of warm and saline Kuroshio water into inner sea (e.g. Kaneda et al., 2002).

The straits may have an important role in the upward transport of nutrients, which generally intrude into the bottom layer or are regenerated at the bed. Fig. 4a and 6a show that, in the euphotic layer, water around straits was typically colder than

that of basins and had relatively high concentration of nutrients. Simpson and Tett (1986) suggested that horizontally inhomogeneous mixing in broad and stratified shelf seas could create a series of frontal regions with a consequent enhancement of biological productivity. The highest concentrations of depth mean chlorophyll (Fig. 7b) was observed in straits S4, S5 which are located on both sides of basin B4. It would appear that the high nutrient levels in B4 leak into the frontal regions between the basin and the adjacent straits and promote growth of phytoplankton. In this way, the longitudinal distribution of mixing intensity created by multiple straits in the Seto Inland Sea seems to induce favorable condition for growth of phytoplankton in large area of Seto Inland Sea.

It is important to remember, of course, that Chlorophyll does not equal primary production but is a rough measure of phytoplankton biomass. Although the linking between tidal stirring and the distribution of the primary producers via water column stability needs to be more fully elucidated in future studies, the present survey results provide persuasive evidence of its importance in the Seto Inland Sea.

In this study, we have used a 3D model with uniform density to simulate the barotropic currents and thus provide new insights into the flow of tidal energy into the Seto Inland Sea system. We recognize, however, that the strong vertical and horizontal gradients observed in the surveys imply the existence of significant baroclinic currents. The simulation of these currents, which are typically 1–2 orders of magnitude smaller than the tidal streams, will require a full 3D density advecting model in future studies. We note also that the rapid variations in mixing rates and vertical structure, which we have observed in the Inland Sea imply that future surveys should aim for improved horizontal resolution with a sampling interval of ~5 km or less in the high gradient regions.

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